M+ terminal, BMS P+
M− terminal, BMS P−
M+ terminal, HS50 +
M− terminal, HS50 −
12 V to Jetson
12 V to Pi 5 V buck
12 V to fans
Pi: 3V3, SDA, SCL, GND, E−stop pressed (low)
L arm +
GND stud 1: L arm, R arm, waist returns
R arm +
waist +
aux +
GND stud 2: aux, L leg, R leg returns
L leg +
R leg +
SPARE_SW, off with the motors
SPARE_ON, always on

1. Battery in and E−stop bank B (LM5069−1, two FETs). Bank A, in series ahead of it, is block 7

The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it.
Rs 0.5 mOhm: limit 97−123 A (VCL 48.5−61.5 mV), breaker 160−260 A (VCB 80−130 mV). Above peak draw and JK DOC 80 A. 1.8 W at 60 A.
Start: Cdv 100 nF, gate 10−22 uA, 100−220 V/s, 0.25−0.55 s to full. Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480−810 W with tolerance: RPWR = 1.3e5 x Rs x (P − 1.18 mV x VDS / Rs), SNVS452G eq 10, VSNS 6 mV over the 5 mV floor. Timer 4.7 nF: 0.22 ms typ, 0.15−0.39 ms.
Fault check: 810 W is 14.8 A at 54.6 V for up to 0.39 ms. Reviewers read about 16 A at 0.82 ms, 70 C, one FET; TI scales as t^0.346, so about 20.7 A at 0.39 ms: 1.4 x. Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 5.2 W hot at 60 A, 1.3 W at 30 A, the same again in bank A. UVLO low sinks the gate at 2 mA: about 170 us off.
The 2 mA also pulls VBUS down through U1's OUT−GATE zener, against the 54 uA of its internal 1 MOhm SENSE−OUT path, so VBUS reads near 0 V with the banks open.
Body diodes of both banks pass regen back to the pack with the banks open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069−1 latches on a fault. Clear: E−stop cycle, then reset.
BAT spike: the absorber in block 10 takes the lead current when a bank opens fast (breaker trip 1.2 us), 5.0SMDJ58A is the backstop. FETs and U1 are 100 V.

Block 10. Absorber (B1): a fast breaker trip leaves the pack lead's current with nowhere to go. The diode lets BAT dump it into 47 uF, so a 200 A, 1 uH lead (20 mJ) charges it about 7 V, plus 6 V across a 30 mOhm ESR at 200 A: about 70 V peak. Polymer, ESR 30 mOhm or less. Measure the lead and scope a 60 A stop. 100k bleeds it. The cap charges through the diode when the BMS closes: 47 uF, 0.07 J, check the BMS takes it.
F1, 1 uF, BAV19W and SMAJ15A (M14, M6, M7): a shorted U2 puts the pack on VCTL; the TVS clamps under the 40 V parts and F1 opens. The BAV19W keeps C5 from back−driving U2 if BAT collapses.
RUN_A, RUN_B and FLT_A diodes (B3): a U1 pin 2−3 bridge or an EN fault can no longer drive RUN, so one fault loses one bank at most.
1 uF on RST_SRC (M8): a reset held at power−up no longer arms the latch (tools/latch_sim.py, 36 cases). Reset interlock (M1): Q10 shorts RST_SRC until MID is over about 11 V through the 10 V zener and Q11, so with bank B shorted and the bus down, reset does nothing; healthy MID is 17 V at a 36 V pack. It loads MID by about 30 uA. 33k on U17 VBUS (M13). 22 nF SRC to GND, TI's recommendation for U8. BAT46W (M4). 330 uF on +12V: spare−output shorts and servo back−feed (M9) land in it and the compute loads.

2. Control rail and E−stop latch (no logic IC)

E−stop: 1 NC out, 2 GND, 3 NC return, 4 GND, 5−6 reset NO, 7 GND, 8 button NO

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

Seal−in latch, ngspice checked at hFE 80: stays off at power−up even with reset held, reset closes it, a 50 us contact bounce rides through,
loop open drops it in about 100 us, re−closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
RUN 7 V. UVLO and EN_A each take RUN through their own diode (block 10), so a fault on either pin cannot drive RUN. UVLO 3.8 V; needs about 4.8 V on RUN_B rising. Loses VCTL, RUN drops, UVLO drops: bank opens.
Loop 6 mA, wetting for a control−rated NC contact. No 12 V on the connector without a series R. GND between loop out and return, and between return and reset: a short there opens the loop or does nothing.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts.
Button HB2−BS545, 22 mm mushroom, twist release, 1NC + 1NO, 660 V AC 10 A, control−rated only: NC in this loop, NO to the Pi (block 9). RUN also enables bank A (block 7) and SPARE_SW (block 8).

3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into the HS50

Arcol HS50 10R J on the frame plate: 322 W, 5.7 A while on. 50 W on Arcol's standard heatsink, 32 g: about 1.4 kJ, 4.5 s at 322 W, before it gets hot. Threshold values are rev A's, unchanged.
Self−powered: VCTL lives as long as the bus or the pack does. If the FET sticks on, the plate thermostat opens the loop and the bank drops the bus.
TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. Rev A's driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.
Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

4. Bus capacitors

400 uF, 80 V. The drives have their own. Can is 10 x 16.5: land owed, and it only clears under the higher ceiling, away from the fans.

5. 12 V brick

Ahead of the E−stop: motors off, compute stays up. Soft start in block 5a, BRK_PG on R.C keeps the brick off until its input is up.
−Vout is tied to GND here: the robot has one ground, P−. RS+ and RS− tied at the pins, Trim open. Input caps at the pins; the EMC filter is left out.
150 W at 12 V, 89 %: 18 W at full load, about 6 W at 50 W. Derating starts near 30 C at 20 LFM: fit a heatsink on the baseplate, in the Z 67 strip, see REV_B_PARTS. 12.7 mm tall: keep clear of the fan gap.
INA228 addresses: L arm 0x40, R arm 0x41, waist 0x42, aux 0x43, L leg 0x44, R leg 0x45, 12 V 0x46, SPARE_SW 0x47, MID 0x48. Powered by the Pi's 3.3 V; Pi pull−ups on I2C.

5a. Brick soft start (ahead of the E−stop switch)

Same controller as the bank, so one part family. −2 retries on its own: a compute fault does not need a BMS cycle. UVLO on VIN, OVLO low: on whenever the pack is.
Power−limit start, no dV/dt network: 70 W limit (Rs 5 mOhm, same equation as U1), about 1.3 A at full pack. 4.4 uF plus the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 47 nF: 2.2 ms typ, 1.5−3.8 ms. Limit 9.7−12.3 A clears the worst brick draw, 4.7 A at 36 V. Brick input capacitance is unpublished: about 40 uF is the most the minimum timer covers, so measure one brick.
Fault check: 87 W is 1.6 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)

9. E−stop pressed: button NO contact to the Pi

Low means pressed. Pi GPIO input, no Pi pull−up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.
A harness short to the 54.6 V bus: the BAT46W blocks the pull−up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.
Information only: the NC loop does the stopping.

8a. SPARE_ON eFuse

Unswitched, about 2 A limit. Same part as U9: a short stops in 5 us and leaves compute up.

6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.

1 mOhm: 60 A peak is 60 mV, ADCRANGE 0. 0.36 W at 19 A.
Legs DNP: fit the same parts when they arrive.
Returns share two GND studs, three limbs each: check the stud and its lug for the summed current.

7. E−stop bank A: second series bank, own driver, ahead of bank B

Own driver: a FET usually dies shorted gate to drain, which would pull a shared gate net to the pack and hold the other bank on. RUN is the only shared signal, and each bank takes it through its own diode.
Both banks face the same way, drains toward the pack, so regen stays open. Bank B keeps U1, the shunt and the soft start.
Close order: U8 FLT holds U1 UVLO low through a 100 V diode until BST is up (about 57 ms typ, 35−115 ms with 2.2 uF). The diode keeps a UVLO fault off FLT (20 V max), so A closes first by logic, not by timing. A closes onto MID only. FLT also blocks B if U8 has no charge pump or has tripped.
Off edge: EN falls first, 4.6 us typ, 6.5 us max, then G1PD pulls 390 nC through 150R: about 10 us, so 6 A/us at 60 A, against a 100 us latch. D1 stays as the backstop. Check at the bench.
Short−circuit comparator on RS1: 40.2k gives 99 mV, 79−121 mV, 158−242 A, above U1's limit. TMR 100k + 680 pF: about 10 us, latch−off, cleared by an E−stop cycle. It covers reset with B shorted: A opens, and FLT holds B off.
Driver off holds gate to source with no path to GND, so MID can be biased: 47k + 47k, 0.48−0.50 x pack, 0.6 mA, 63 mW each. Hot FET leakage, up to 200 uA per bank, moves MID by under 5 V. U17 reads MID through 33k (spike and fault limit): calibrate the 3 % gain.
Self−test before reset, pack 36−54.6 V, BMS telemetry for pack voltage: MID 0.4−0.6 x pack and U11 VBUS under 3 V is healthy. MID within 2 V of pack: A shorted. MID under 3 V: B shorted. VBUS high: both shorted, or the bus has not bled down yet, about 11 s on board caps, longer with drives.
U17 uses VBUS only, IN+ and IN− on GND. VBUS alone cannot see one shorted bank, which is why MID is read.

8. Spare 12 V outputs

SPARE_SW: RUN enables it, so it drops with the motor bank. The divider keeps 7.7 V of RUN under the 6 V EN limit. Limit 3.87−4.42 A (487R). dV/dt 3.3 nF: about 12 V/ms. One switch: a shorted U9 stays on. Current and volts at 0x47, 5 mOhm, ADCRANGE 1.
SPARE_ON: block 8a. Both spares sit behind RS8, so the 12 V meter counts them.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR
marengo
Sheet: /
File: marengo_pdu_revb.kicad_sch
Title: marengo power board rev B
Size: A1 | Date: 2026−11−06 | **Rev: B−wip**
KiCad E.D.A. 10.0.8 | Id: 1/1